Correlation

Correlation helps to check whether there exist any relationship between two varibles and if it exist then to what extent these variables are related to each other.

Correlation can be classified into two types

1) Positive Correlation

If increment (or decrement) in one variable causes the increment (or decrement) in other variable, then it is said to be positive correlation.
(in other words both variables changes in same direction)

e.g. – Income & expenditure.

2) Negative Correlation

If increment (or decrement) in one variable causes the decrement (or increment) in other variable, then it is said to be negative correlation.
(in other words both variables changes in opposite direction)

e.g. – Price & sale of a product

Karl Pearson's coefficient of correlation

The formula for coefficient of correlation was developed by Karl-Pearson.

It is used to measure the intensity / extent of a linear relationship between two variables.

Karl-Pearson's coeff^n of correlation or simply correlation coeff^n between two variables x & y is denoted and defined by

$$\boxed{r(x,y) = r = \frac{cov(x,y)}{\sigma_x \cdot \sigma_y}}$$

where

$$Cov(x,y) = \text{Covariance of } x \text{ \& } y$$

$$= \frac{1}{n}\sum(x-\bar{x})(y-\bar{y})$$

$$= \frac{1}{n}\sum[xy - x\bar{y} - \bar{x}y + \bar{x}\bar{y}]$$

$\frac{\sum x}{n} = \bar{x}$

$$= \frac{\sum xy}{n} - \bar{y}\frac{\sum x}{n} - \bar{x}\frac{\sum y}{n} + \bar{x}\bar{y}\frac{\sum 1}{n}$$

$\frac{\sum y}{n} = \bar{y}$

$$= \frac{\sum xy}{n} - \bar{y}\bar{x} - \cancel{\bar{x}}\bar{y} + \cancel{\bar{x}}\bar{y}$$

$\sum 1 = n$

$$= \frac{\sum xy}{n} - \bar{x}\bar{y}$$

Thus

$$\boxed{cov(x,y) = \frac{1}{n}\sum(x-\bar{x})(y-\bar{y}) = \frac{1}{n}\sum xy - \bar{x}\bar{y}}$$

σ_x – Standard deviation in x

$$= \sqrt{\frac{1}{n}\Sigma(x-\bar{x})^2} = \sqrt{\frac{1}{n}\Sigma x^2 - \bar{x}^2}$$

σ_y – Standard deviation in y

$$= \sqrt{\frac{1}{n}\Sigma(y-\bar{y})^2} = \sqrt{\frac{1}{n}\Sigma y^2 - \bar{y}^2}$$

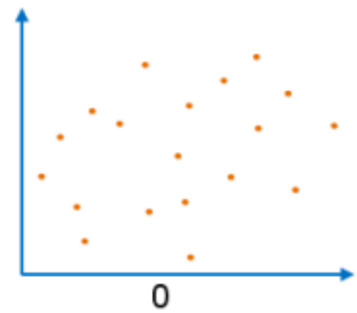
Substituting these vallues in r, we get

$$\boxed{\begin{aligned} r &= \frac{cov(x, y)}{\sigma_x \, \sigma_y} \\ &= \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\sqrt{\Sigma(x-\bar{x})^2 \cdot \Sigma(y-\bar{y})^2}} \\ &= \frac{\frac{1}{n}\Sigma xy - \bar{x}\cdot\bar{y}}{\sqrt{\left(\frac{1}{n}\Sigma x^2 - \bar{x}^2\right)\left(\frac{1}{n}\Sigma y^2 - \bar{y}^2\right)}} \end{aligned}}$$

Properties of correlation coefficient

1) Value of r always lies in between -1 to $+1$ i.e.

$$-1 < r < 1$$

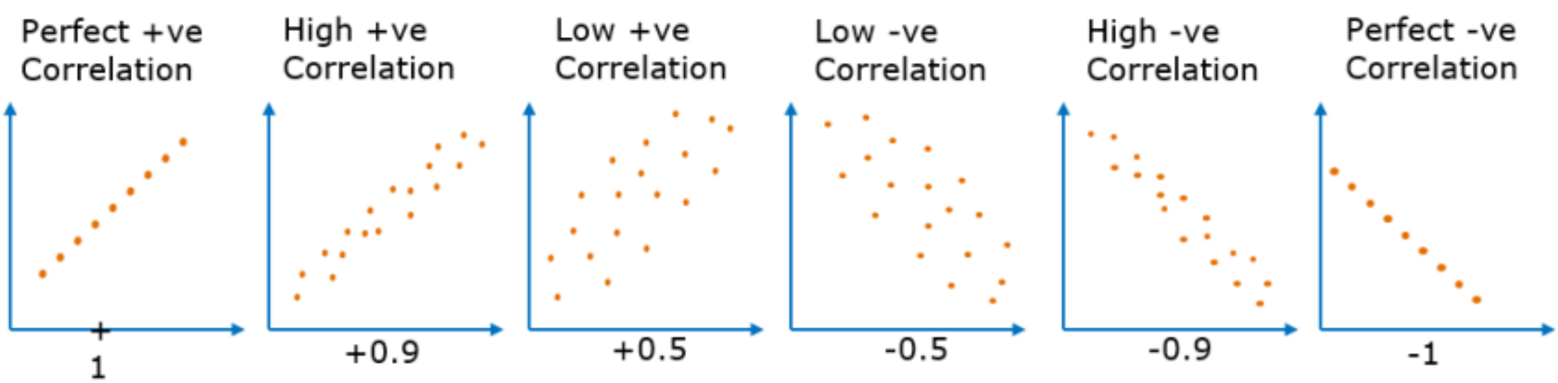


$r = 0 \Rightarrow$ No correlation

$r = \pm 1 \Rightarrow$ Perfect correlation

$r > 0 \Rightarrow$ Positive correlation

$r < 0 \Rightarrow$ Negative correlation



2)

Value of r	Correlation
$-0.39 < r < 0.39$	Weak correlation
$-0.79 < r < -0.4$ $0.4 < r < 0.79$	Moderate correlation
$-1 < r < -0.8$ $0.8 < r < 1$	Strong correlation

3) Correlation coeff^n is independent of shift of origin as well as change of scale.

i.e. if we put $u = \frac{x-A}{h}$ & $v = \frac{y-B}{k}$

where A, B, h, k are all constants. Then

$$\boxed{r(x, y) = r(u, v)}$$

1. Find the coefficient of correlation for the following data

x	10	14	18	22	26	30
y	18	12	24	06	30	36

Sol^n :-

	x	y	x^2	y^2	x*y
	10	18	100	324	180
	14	12	196	144	168
	18	24	324	576	432
	22	6	484	36	132
	26	30	676	900	780
	30	36	900	1296	1080
Total =	120	126	2680	3276	2772

Here $n = 6$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{120}{6} = 20$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{126}{6} = 21$$

We have

$$r = \frac{\frac{1}{n}\sum xy - \bar{x}\cdot\bar{y}}{\sqrt{\left(\frac{1}{n}\sum x^2 - \bar{x}^2\right)\left(\frac{1}{n}\sum y^2 - \bar{y}^2\right)}}$$

$$= \frac{\frac{2772}{6} - (20\times 21)}{\sqrt{\left(\frac{2680}{6} - 20^2\right)\left(\frac{3276}{6} - 21^2\right)}}$$

$$= 0.6$$

$$\boxed{r = 0.6}$$

There exist moderate correlation in x & y

(2) Calculate the coefficient of correlation for the following data.

X	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

Solⁿ

	x	y	x^2	y^2	x*y
	1	9	1	81	9
	2	8	4	64	16
	3	10	9	100	30
	4	12	16	144	48
	5	11	25	121	55
	6	13	36	169	78
	7	14	49	196	98
	8	16	64	256	128
	9	15	81	225	135
Total =	45	108	285	1356	597

Here $n = 9$

$$\bar{x} = \frac{\sum x}{n} = \frac{45}{9} = 5$$

$$\bar{y} = \frac{\sum y}{n} = \frac{108}{9} = 18$$

We have

$$r = \frac{\frac{1}{n}\Sigma xy - \bar{x}\cdot\bar{y}}{\sqrt{\left(\frac{1}{n}\Sigma x^2 - \bar{x}^2\right)\left(\frac{1}{n}\Sigma y^2 - \bar{y}^2\right)}}$$

$$= \frac{\frac{597}{9} - (5\times 18)}{\sqrt{\left(\frac{285}{9} - 5^2\right)\left(\frac{1356}{9} - 18^2\right)}}$$

$$= 0.95$$

$$\boxed{r = 0.95} \Rightarrow \text{Strong positive correlation}$$

3 Calculate the correlation coefficient for the following data

$n = 20, \sum x = 40, \sum y = 40, \sum x^2 = 190, \sum y^2 = 200, \sum xy = 150.$

Soln:

$$\bar{x} = \frac{\Sigma x}{n} = \frac{40}{20} = 2$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{40}{20} = 2$$

We have

$$r = \frac{\frac{1}{n}\Sigma xy - \bar{x}\cdot\bar{y}}{\sqrt{\left(\frac{1}{n}\Sigma x^2 - \bar{x}^2\right)\left(\frac{1}{n}\Sigma y^2 - \bar{y}^2\right)}}$$

$$= \frac{\frac{150}{20} - (2\times 2)}{\sqrt{\left(\frac{190}{20} - 2^2\right)\left(\frac{200}{20} - 2^2\right)}} = 0.6092$$

$$\boxed{r = 0.6092}$$

4. Calculate the coefficient of correlation for the following data.

x	61	64	67	70	73
y	5	18	42	27	8

Soln:-

	x	y	x^2	y^2	x*y
	61	5	3721	25	305
	64	18	4096	324	1152
	67	42	4489	1764	2814
	70	27	4900	729	1890
	73	8	5329	64	584
Total	335	100	22535	2906	6745

Here $n = 5$

$$\bar{x} = \frac{\sum x}{n} = \frac{335}{5} = 67$$

$$\bar{y} = \frac{\sum y}{n} = \frac{100}{5} = 20$$

We have

$$r = \frac{\frac{1}{n}\sum xy - \bar{x}\cdot\bar{y}}{\sqrt{\left(\frac{1}{n}\sum x^2 - \bar{x}^2\right)\left(\frac{1}{n}\sum y^2 - \bar{y}^2\right)}}$$

$$= \frac{\frac{6745}{5} - (67 \times 20)}{\sqrt{\left(\frac{22535}{5} - 67^2\right)\left(\frac{2906}{5} - 20^2\right)}}$$

$$= 0.1575$$

$\boxed{r = 0.1575}$ $\Rightarrow$ Weak correlation

⑤ Calculate the coefficient of correlation for the following data.

X	10	14	19	26	30	34
Y	12	16	18	26	29	35

Solⁿ :-

	x	y	x^2	y^2	x*y
	10	12	100	144	120
	14	16	196	256	224
	19	18	361	324	342
	26	26	676	676	676
	30	29	900	841	870
	34	35	1156	1225	1190
Total	**133**	**136**	**3389**	**3466**	**3422**

Here $n = 6$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{133}{6} = 22.1667$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{136}{6} = 22.6667$$

We have

$$r = \frac{\frac{1}{n}\Sigma xy - \bar{x}\cdot\bar{y}}{\sqrt{\left(\frac{1}{n}\Sigma x^2 - \bar{x}^2\right)\left(\frac{1}{n}\Sigma y^2 - \bar{y}^2\right)}}$$

$$= \frac{\frac{3422}{6} - (22.1667 \times 22.6667)}{\sqrt{\left(\frac{3389}{6} - (22.1667)^2\right)\left(\frac{3466}{6} - 22.6667^2\right)}}$$

$$= 0.9909$$

$\boxed{r = 0.9909}$ $\Rightarrow$ strong correlation

⑥ Obtain correlation coefficient for the following data.

x	200	300	400	500	700
y	2	10	6	8	2

Here we will use shift of origin and change of scale property

Let $u = \frac{x-400}{100}$

then $r(x,y) = r(u,y)$

$$\therefore\ r(x,y) = r(u,y) = \frac{\frac{1}{n}\Sigma u \cdot y - \bar{u}\,\bar{y}}{\sqrt{\left(\frac{\Sigma u^2}{n} - \bar{u}^2\right)\left(\frac{\Sigma y^2}{n} - \bar{y}^2\right)}}$$

	x	y	u	u^2	y^2	u*y
	200	12	-2	4	144	-24
	300	10	-1	1	100	-10
	400	6	0	0	36	0
	500	8	1	1	64	8
	700	2	3	9	4	6
Total	2100	38	1	15	348	-20

Here $n = 5$

$$\bar{u} = \frac{\Sigma u}{n} = \frac{1}{5} = 0.2$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{38}{5} = 7.5$$

$$\therefore\ r(x,y) = r(u,y) = \frac{\frac{-20}{5} - (0.2 \times 7.5)}{\sqrt{\left(\frac{15}{5} - 0.2^2\right)\left(\frac{348}{5} - 7.5^2\right)}}$$

$$= -0.9324$$

$\boxed{r = -0.9324}$ => Strong negative correlaⁿ